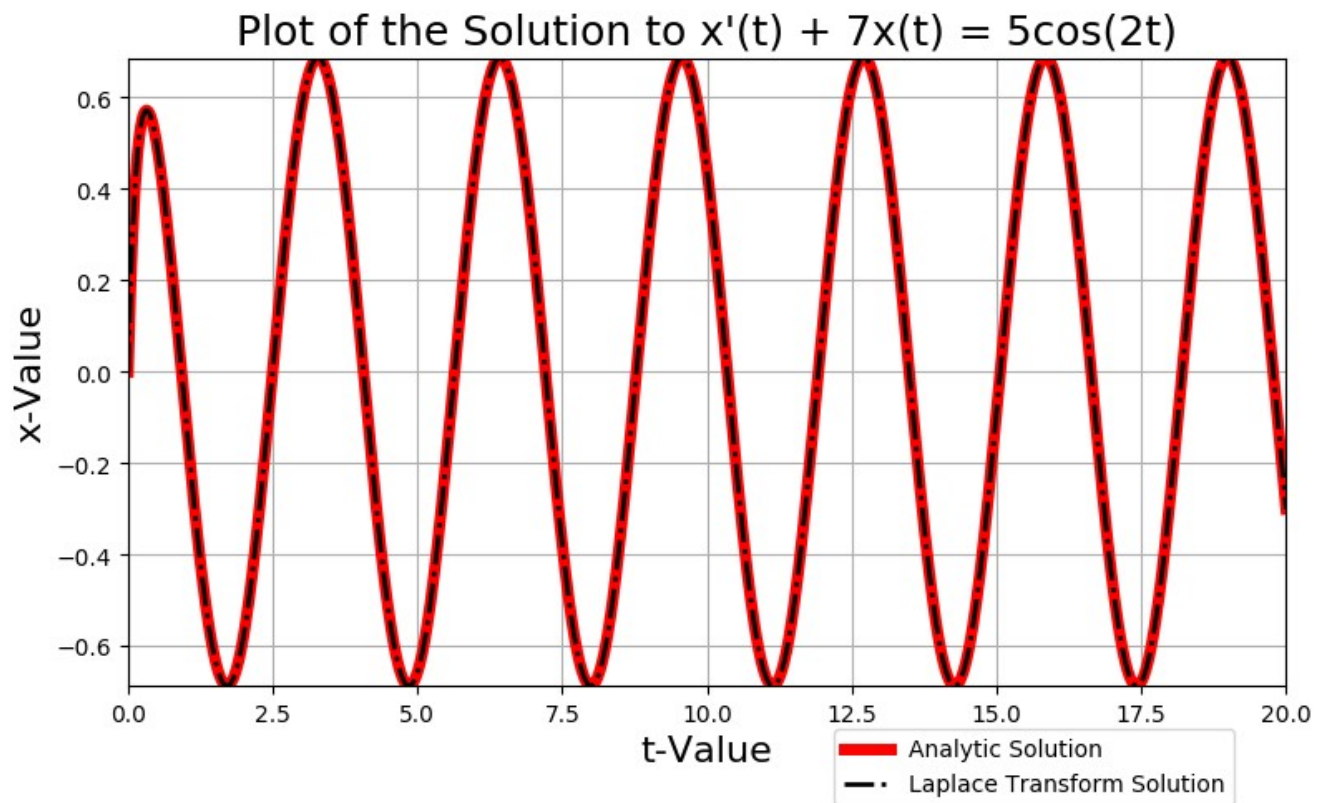


In [7]: runfile('C:/Anaconda3/My_Scripts/Controls/ME_426_HW_4.py', wdir='C:/Anaconda3/My_Scripts/Controls')

Problem 20a)

Transfer Function of $x'(t) + 7x(t) = 5\cos(2t)$:

$$\frac{5s}{s^3 + 7s^2 + 4s + 28}$$

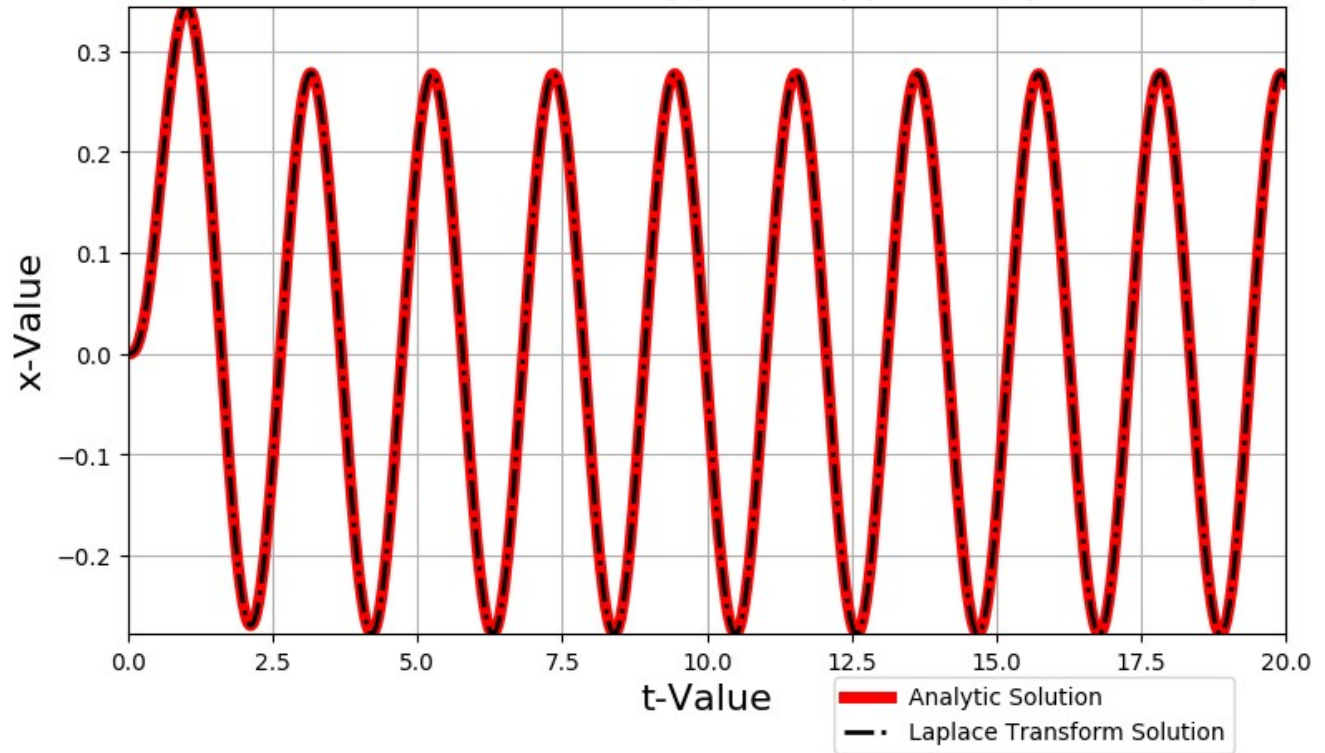


Problem 20b)

Transfer Function of $x''(t) + 6x'(t) + 8x(t) = 5\sin(3t)$:

$$\frac{15}{s^4 + 6s^3 + 17s^2 + 54s + 72}$$

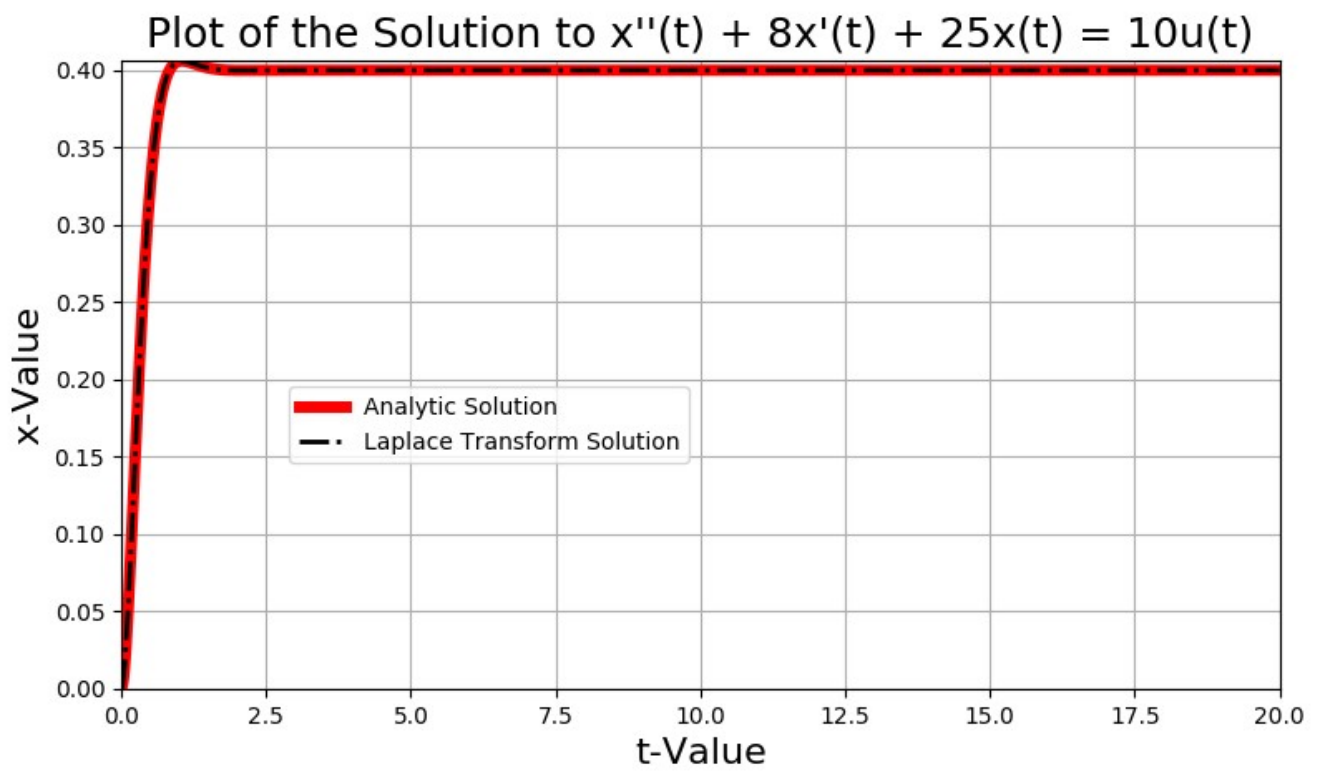
Plot of the Solution to $x''(t) + 6x'(t) + 8x(t) = 5\sin(3t)$



Problem 20c)

Transfer Function of $x''(t) + 8x'(t) + 25x(t) = 10u(t)$:

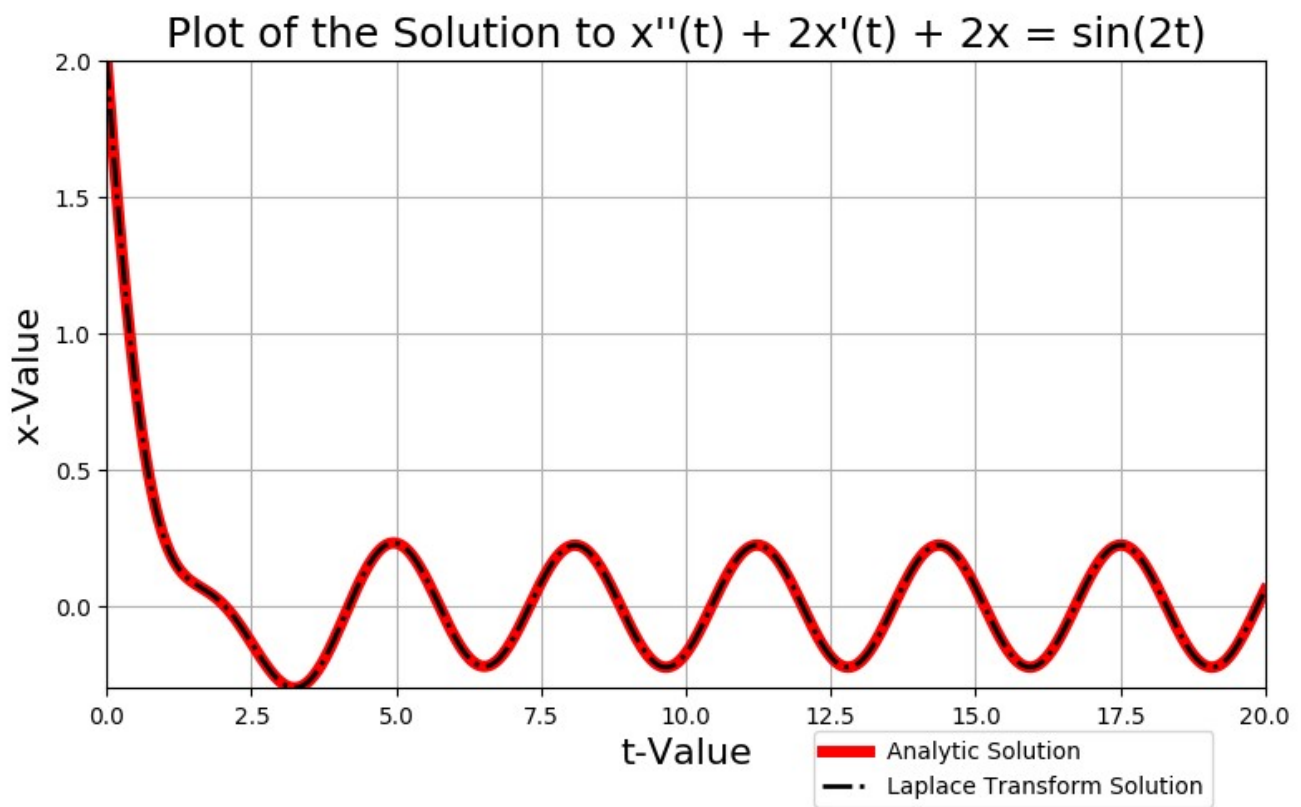
$$\frac{10}{s^3 + 8s^2 + 25s}$$



Problem 21a)

Transfer Function of $x''(t) + 2x'(t) + 2x = \sin(2t)$:

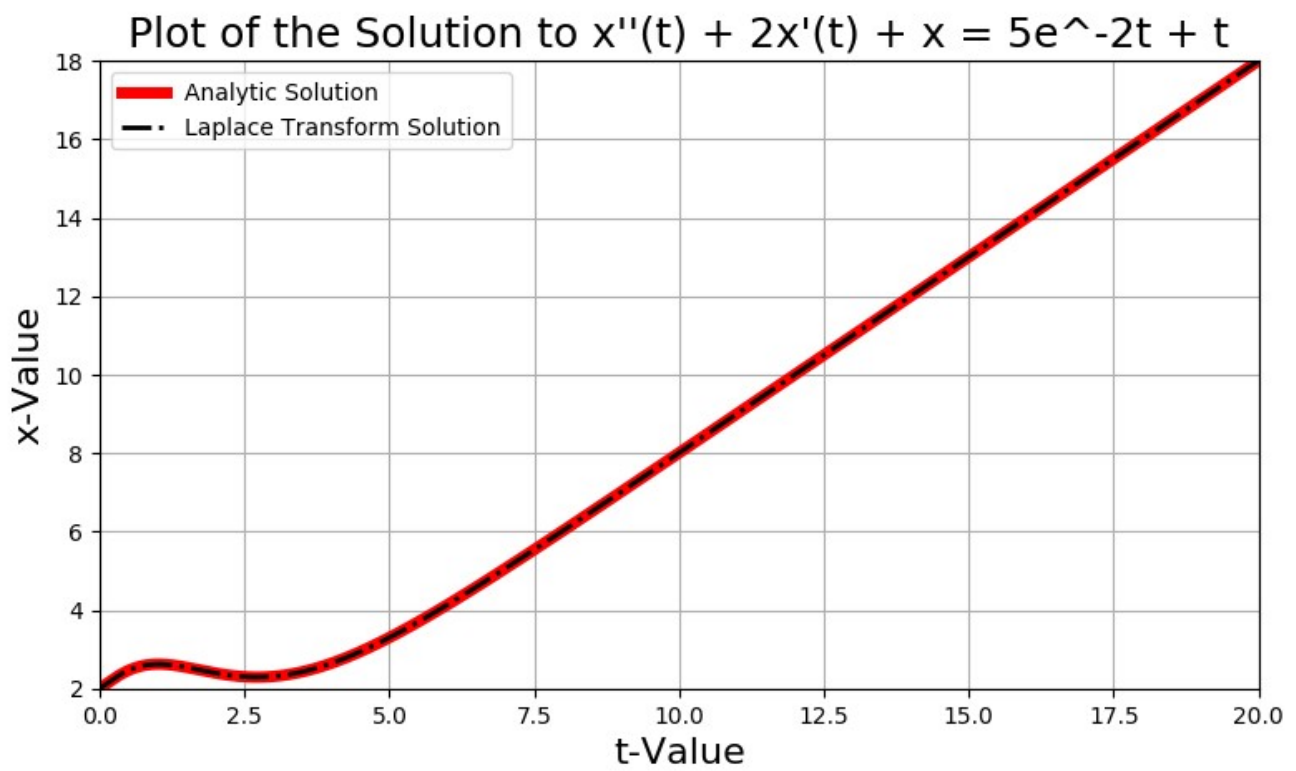
$$\frac{2s^3 + s^2 + 8s + 6}{s^4 + 2s^3 + 6s^2 + 8s + 8}$$



Problem 21b)

Transfer Function of $x''(t) + 2x'(t) + x = 5e^{-2t} + t$:

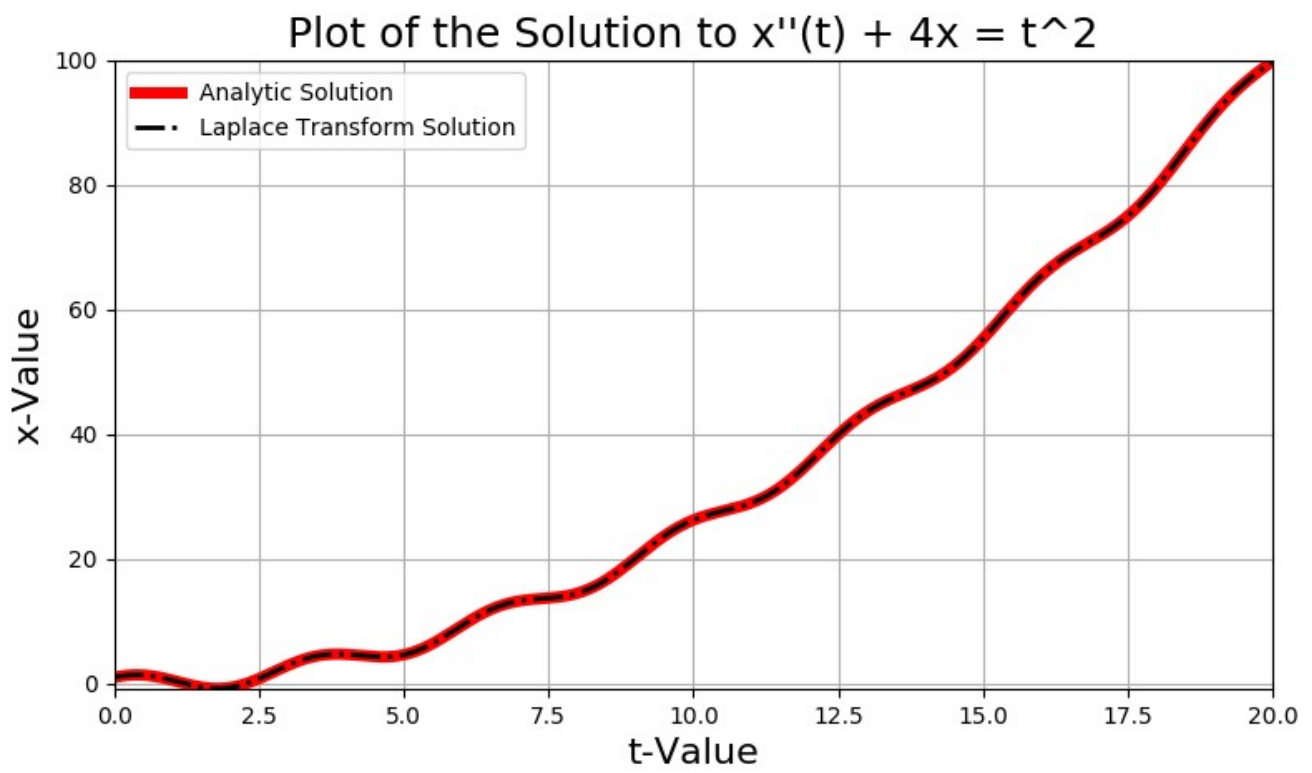
$$\frac{2s^4 + 9s^3 + 15s^2 + s + 2}{s^5 + 4s^4 + 5s^3 + 2s^2}$$



Problem 21c)

Transfer Function of $x''(t) + 4x = t^2$:

$$\frac{s^4 + 2s^3 + 2}{s^5 + 4s^3}$$



In [8]: